Building a daily English tutor inside OpenClaw for Telegram

The most effective approach combines pre-generated structured lessons with lightweight LLM calls for personalization — a hybrid architecture that fits comfortably within Gemini's free tier while delivering genuine A1-level progression. Existing Telegram language bots are either dictionary-based or primitive chatbots; none combine LLM-powered adaptive teaching with spaced repetition and a structured curriculum. This gap is your opportunity. The OpenClaw skill system, built around a `SKILL.md` file (Openclaw) with cron-based scheduling and native Telegram delivery, provides the exact infrastructure needed. (Apiyi.com Blog) With **2–5 Gemini API calls per day** (well under even the most restrictive free-tier limits), you can deliver daily 7–10 minute lessons that outperform every current Telegram English bot. (AI Free API)

What OpenClaw skills look like under the hood

OpenClaw skills follow the AgentSkills specification. A skill is a directory with one required file: **`SKILL.md`** (not `skill.json` — the platform uses Markdown with YAML frontmatter, not JSON). (Smithery) The minimal structure looks like this:

```
english-tutor/
├── SKILL.md            # Required: YAML frontmatter + instructions
├── scripts/            # Optional: Python/Bash helpers
│   └── generate-lesson.py
├── references/         # Optional: loaded into context on demand
│   └── curriculum.md
├── assets/             # Optional: templates, word lists
│   └── vocabulary.json
└── config.json         # Optional: config schema
```

The `SKILL.md` frontmatter defines the skill's identity and requirements:

```yaml
---
name: english-daily-tutor
description: Generate and deliver a daily English lesson for a Russian-speaking beginner via Telegram
user-invocable: true
metadata: {"openclaw":{"emoji":"🇬🇧","requires":{"env":["GEMINI_API_KEY"]},"primaryEnv":"GEMINI_API_KE
---
```

Below the frontmatter, you write natural-language instructions that tell the AI agent exactly how to execute the skill — what to generate, how to format it, where to send it. Lumadock Zenvanriel

Cron scheduling runs inside OpenClaw's Gateway process and persists across restarts. Openclaw openclaw For daily lessons, you'd configure an isolated session job:

```json
{
  "name": "Daily English Lesson",
  "schedule": { "kind": "cron", "expr": "0 9 * * *", "tz": "Europe/Moscow" },
  "sessionTarget": "isolated",
  "payload": {
    "kind": "agentTurn",
    "message": "Generate today's English lesson and deliver it."
  },
  "delivery": {
    "mode": "announce",
    "channel": "telegram",
    "to": "-100XXXXXXXXX:topic:YYY"
  }
}
```

Isolated sessions are critical here — they prevent lesson generation from polluting your main chat context. Zenvanriel openclaw The delivery target supports direct chat IDs, group chats, and forum topics. One practical note: single-user setups currently require explicit `--channel telegram --reply-to <chat_id>` on each job (a feature request for `defaultTo` config exists as Issue #15925). GitHub

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The landscape of English learning for Russian speakers

Russian speakers face a distinctive set of challenges that generic English courses handle poorly. **Articles (a/an/the) are the single most persistent error** — Russian has no articles at all, so learners systematically omit them ("I am good person" instead of "I am a good person"). Study The verb "to be" is dropped in Russian present tense ("She teacher" is grammatically correct in Russian), causing English sentences like "I doctor" instead of "I am a doctor." TALK Schools Other major pain points include the **/θ/ and /ð/ sounds** (Russian substitutes /s/, /t/, or /f/), TEFL Course the distinction between /w/ and /v/ (nonexistent in Russian), Heather Hughes and the English tense system — Russian has 3 tenses versus English's 12+, with Present Perfect being especially alien. Heather Hughes

The best existing products address these challenges in different ways. **Duolingo** succeeds at habit formation through gamification (streaks, XP, leagues) but fails at grammar explanation Duolingo — its implicit "learn by doing" approach frustrates LingoDeer Russian speakers who expect and benefit from explicit rules. **Puzzle English**, designed specifically for Russian speakers, excels at listening comprehension through video/audio puzzles and uses the SuperMemo algorithm for vocabulary. Tenereteam **LinguaLeo** (23 million users) App Store pioneered content-based learning where users extract vocabulary from authentic materials App Store like TED Talks. **Skyeng's free AI Tutor bot** on Telegram represents the current state of the art for Russian-market AI language bots, offering voice conversation practice with pronunciation feedback and CEFR scoring. Skysmart

Research consistently shows that **explicit grammar instruction in Russian** outperforms immersive English-only approaches for beginners. Academia.edu The Russian educational tradition is grammar-focused, and the structural gulf between Russian and English (articles, word order, auxiliary verbs) demands direct explanation. TALK Schools Contrastive analysis — explicitly comparing Russian and English structures — accelerates understanding. Wosjournals

Among Telegram bots specifically, the landscape is surprisingly weak. A systematic test of 23 English-learning bots found **12 were already broken**, Habr and the most famous bot (@AndyRobot) was dismissed as "a toy for 15 minutes" that becomes repetitive and topic-jumpy. ITMO News habr The best-rated multi-functional bot (@EnglishSimpleBot) offers reading, word saving, and grammar exercises but lacks AI-powered adaptivity. Habr @Wordich uses Ebbinghaus-curve spaced repetition with 10 daily words and gets positive reviews for fitting into morning routines. Wordich The critical gap: **no existing Telegram bot combines LLM-powered personalization with structured curriculum progression and spaced repetition.**

How AI tutors teach languages and what actually works

The AI language learning space has matured rapidly. **Praktika** (10M+ downloads, Google Play strong Russian market presence) uses lifelike avatars with 0.1-second response times ICanLearn and a 5-step learning flow: learn concepts in your native language, build a study plan, get smart prompts, receive instant corrections, then speak freely. Praktika **Langua** is considered the best AI speaking app in 2026, featuring spaced-repetition flashcards generated from conversations and AI memory that tracks user interests and recurring mistakes. Lingtuitive **Speak** built proprietary speech recognition specifically for accented speakers — their core insight was that general-purpose ASR fails non-native learners. OpenAI

Research from the CSEDU 2025 conference confirms that LLM-based role-play tutoring reduces foreign language anxiety — students find it easier to make mistakes with a computer than with peers. scitepress A key paper on controlling difficulty introduced the **Token Miss Rate (TMR)** metric, measuring what percentage of AI output exceeds the learner's vocabulary level. The FUDGE method (modular difficulty control) performed best at keeping output within learner-appropriate bounds without sacrificing fluency. arXiv

The most effective prompting strategies for language teaching follow a clear hierarchy:

- **Recasting over explicit correction**: Repeat the learner's sentence correctly rather than stopping to explain errors — this maintains conversational flow

- **Scaffolding within the Zone of Proximal Development**: Break complex content into steps, provide real-time support, adapt difficulty dynamically (arXiv)
- **Target a 60–70% success rate**: This is the optimal difficulty zone — challenging enough to learn, easy enough to avoid frustration (GitHub)
- **Persona-based system prompts**: Define the tutor's personality, correction style, language-mixing rules, and difficulty constraints in a concise system prompt (arXiv)
- **Structured JSON output**: Force concise, parseable responses that eliminate wasted tokens on narrative text (Data Studios)

A 90-day A1 curriculum built for daily Telegram delivery

The **New General Service List's first 560 words** provide ~92% coverage of general English text (Grokipedia) and form the vocabulary backbone. Combined with the empirically tested grammar progression from the Oxford Navigate A1 coursebook, (english-malta) here's how to structure the curriculum:

Weeks 1–2 (Foundations): Verb "to be" with articles — tackling the two biggest Russian-speaker errors simultaneously. Subject pronouns, numbers 1–20, colors. ~70 new words. (english-malta) Every lesson hammers "I am a..." and "She is a..." patterns.

Weeks 3–4 (Personal World): Family, countries, jobs, possessive adjectives. Introduce "this/that." (english-malta) ~70 new words. Article practice continues as background reinforcement.

Weeks 5–6 (Daily Life): Daily routines, time expressions, days of the week. Present simple with "do/does" questions — auxiliary verbs are alien to Russian grammar and need heavy drilling. ~70 new words.

Weeks 7–8 (Food and Places): Food, drinks, shopping, places in town. Present simple with he/she/it, some/any, countable/uncountable nouns. Preposition focus. ~70 new words.

Weeks 9–10 (Description): Home, rooms, weather, clothes. "There is/are," "can/can't," present continuous (no Russian equivalent — needs careful introduction). ~70 new words.

Weeks 11–12 (Past and Future): Travel, holidays, life events. Past simple with regular and irregular verbs, "going to" for future plans. ~70 new words.

Week 13 (Consolidation): Review all grammar and vocabulary. Total: **~420 vocabulary items + core A1 grammar across 90 days.**

Research consensus points to **5–7 new words per day** as optimal for adult beginners using spaced repetition. (LanGeek) Sessions of **7–10 minutes** hit the sweet spot — short enough for mobile, long enough for depth. (Eduresearchjournal) Each daily lesson should follow a three-phase structure: review (3–4 min of SRS flashcard-style recall covering ~12–16 due items), new content (3–4 min introducing 5–7 words or 1 grammar point with examples), and practice (2–3 min of exercises applying new and reviewed content).

For spaced repetition, a **simplified 5-box Leitner system** is ideal given the constraints. Each vocabulary item tracks just 5 fields: word ID, box number (1–5), last reviewed date, next review date, and times correct. (Wikipedia) Box intervals: daily → every 2 days → every 5 days → every 14 days → every 30 days. Correct answers promote to the next box; incorrect answers return to Box 1. (LanGeek) This requires minimal state management — a single JSON file suffices.

Technical architecture that fits free-tier constraints

Gemini 2.5 Flash-Lite at **1,000 requests per day** on the free tier is the workhorse model. (AI Free API) Even Gemini 2.5 Flash at 250 RPD provides enormous headroom for a single-user app needing 2–5 calls daily. (AI Free API) Token usage per lesson runs approximately **2,000–3,000 tokens total** (input + output), negligible against the 250K TPM limit. (AI Free API)

The key architectural insight is a **hybrid pre-generation + real-time generation** model:

Pre-generated content (zero API calls at delivery time): Vocabulary definitions, translations, and example sentences stored in a local JSON file. Grammar explanations written as static templates (12–15

topics, pre-authored in Russian). Standard exercise formats (fill-in-blank, matching, multiple choice) parameterized from the vocabulary database. This eliminates **70–80% of potential API calls**.

Real-time LLM generation (2–5 calls per lesson): Personalized feedback on user answers (1–2 calls). Adaptive exercise generation based on weak areas (1 call). An occasional fresh example sentence weaving recently learned words together (1 call). A motivational summary or mini-conversation snippet (1 call).

For even greater efficiency, run a **weekly batch pre-generation job** — 7 API calls on Sunday evening to generate the next week's lesson skeletons, cached locally. This reduces daily real-time calls to 1–2 for feedback only.

Gemini's native structured JSON output (`response_mime_type: "application/json"` with `response_schema`) is essential for token efficiency. Prospera Soft Google AI A vocabulary card generation schema would look like:

```json
{
  "type": "array",
  "items": {
    "type": "object",
    "properties": {
      "word": {"type": "string"},
      "translation_ru": {"type": "string"},
      "example_en": {"type": "string"},
      "example_ru": {"type": "string"},
      "part_of_speech": {"type": "string"}
    }
  }
}
```

System prompts should stay under **100–200 tokens**: Portkey "You are an English teacher for a Russian-speaking A1 beginner. Generate CEFR A1-level content. Use only vocabulary from the provided word

list. Respond in JSON only." This concision matters because every token in the system prompt is repeated on every call. Medium

The **$20/month Claude subscription** serves as a fallback for complex tasks — generating nuanced grammar explanations that compare Russian and English structures, or creating contextually rich dialogue scenarios. Reserve Claude for weekly content pre-generation where quality matters most; use Gemini Flash-Lite for daily lightweight tasks.

Concrete recommendations for your OpenClaw skill

The daily lesson delivered via Telegram should follow this format, proven effective by the best-performing Telegram bots:

Message structure: Start with a greeting and day/lesson number (streak reinforcement). Present 2–3 review words as inline flashcards with emoji markers (🔄). Introduce the day's new vocabulary (5–7 words) with Russian translations, phonetic hints for tricky pronunciation, and one example sentence each. Include one **mini grammar note** in Russian explaining a specific English pattern (articles, "to be," word order) with a contrastive Russian example. End with 2–3 quick exercises (fill-in-blank, choose correct, translate) using inline keyboard buttons for answers. Keep total message length under Telegram's 4,096-character limit. Fixmebot

What makes Telegram bots engaging vs. ignored: Inline keyboard buttons dramatically outperform typed commands. Short messages with emoji visual markers (✅ ❌ 🔊 💡) get read; walls of text get skipped. Morning delivery (9 AM local time) fits existing routines — @Wordich users specifically praise "10 minutes before leaving home." Wordich An "I'm stuck / Подсказка" button reduces frustration and prevents abandonment. Skysmart Weekly progress summaries with concrete numbers ("You learned 35 new words this week, 78% retention") maintain motivation.

What to avoid: Don't attempt free-form conversation at the beginner level — @AndyRobot proves this becomes repetitive and frustrating within 15 minutes. ITMO News habr Don't use English-only instruction for A1 learners — Russian explanations are essential. TALK Schools Don't introduce grammar implicitly (the Duolingo trap) — Russian speakers need explicit rules. LingoDeer Don't send lessons

longer than 10 minutes of content — microlearning research shows SP Cloud Academy 8–12 minutes is optimal. Eduresearchjournal

Implementation priority for the SKILL.md: The skill's natural-language instructions should direct the agent to: (1) check the SRS state file for due review items, (2) select the next vocabulary batch from the curriculum JSON, (3) call Gemini to generate exercises incorporating both review and new words, (4) format the lesson as a Telegram message with inline markup and emoji, (5) deliver via the configured chat ID, (6) update the SRS state file. The `references/` directory should contain `curriculum.json` (all 420+ vocabulary items with metadata), `grammar-templates.md` (pre-written A1 grammar explanations in Russian), and `srs-state.json` (Leitner box tracking for each word).

This architecture delivers a product that fills a genuine gap: **the first AI-powered, curriculum-structured, SRS-integrated English tutor on Telegram for Russian speakers** — built entirely on free-tier infrastructure.